Homework 1

September 2026

Due: 2026.10.11 23:59:00

1 Short questions (25 points)

1.1 Homogeneous coordinates (10 points)

A point in N-dimensional space can be represented by Cartesian coordinates or by homogeneous coordinates. The homogeneous representation of a point is not unique.

1. Let $\mathbf{p} = (p_1, \ldots, p_N, s)$ and $\mathbf{q} = (q_1, \ldots, q_N, t)$ be homogeneous coordinate vectors. Their last coordinates, s and t, are the added coordinates, with $s, t \neq 0$. Give a necessary and sufficient condition for $\mathbf{p}$ and $\mathbf{q}$ to represent the same Cartesian point, and write its Cartesian coordinates. (4 points)

2. Explain two advantages of homogeneous coordinates in computer vision. (6 points)

1.2 Rolling shutter (15 points)

A shutter controls when the image sensor collects light. A global shutter exposes all rows at once, whereas a rolling shutter starts their exposures in sequence; motion can therefore make vertical edges appear tilted.

Figure 1 compares a global-shutter image (top) with a rolling-shutter image (bottom) of a moving train. For this question, assume that the camera is stationary and that row exposures start in order, from top to bottom. The train moves horizontally at constant speed, parallel to the image plane. The door of the train stays at a constant depth from the camera.

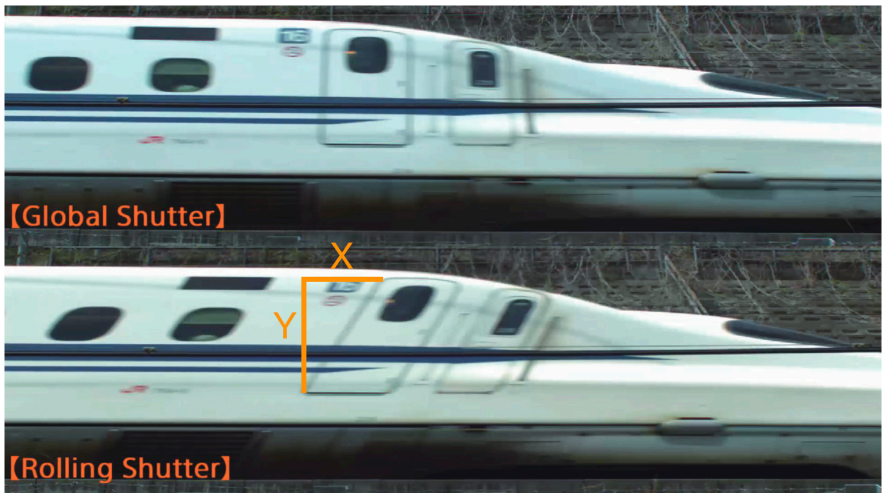


Figure 1: Global and rolling shutters imaging a moving train.

1. From the assumed scan direction, determine whether the train moves left or right in the rolling-shutter image. (3 points)

2. The full rolling-shutter image has $H = 1000$ rows. Scanning all rows takes $T = 30\,\text{ms}$ in total; the time taken to scan each row is the same. Ignore motion blur within each row. The horizontal focal length is $f_x = 6000$ pixels, and the door is at depth $Z = 15\,\text{m}$. In the rolling-shutter image, the door edge spans $Y = 300$ pixels vertically; its endpoints differ by $X = 200$ pixels horizontally (see Figure 1). Calculate the train's speed. Show your formula and calculation steps.[1] (12 points)

2 Camera height from an image (20 points)

Figure 2 shows a 4096×3072 campus image. The ground is planar, and the flowerbed marked in red is 0.76 m high. Point A is on its top edge; let B be the point directly below A on the ground. Please use the full-resolution `campus.png` in the folder for measurements.



Figure 2: Campus image with the flowerbed.

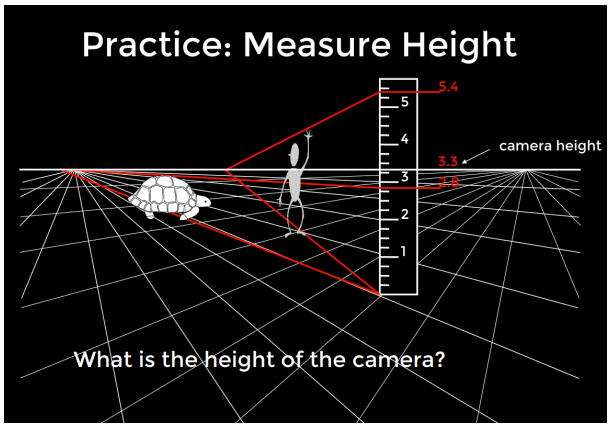


Figure 3: Using a known height and the vanishing line to estimate camera height.

1. Find the vanishing line using two sets of parallel lines. (10 points)

2. Use the reference height to estimate the camera height H. Mark A, B, and the vanishing line; report the image measurements and your calculation. Assume the camera is approximately level and ignore lens distortion. (10 points)

Figure 3 illustrates the height construction.

[1] The numerical values are estimates for this calculation, not real measurements.

3 Image filtering and subsampling (55 points)

This section is designed to help you understand key algorithms in image processing. You will implement these functions from scratch in Python with NumPy:

`cross_correlation_2d`	`convolve_2d`
`gaussian_blur_kernel_2d`	`low_pass`
`image_subsampling`	`gaussian_pyramid`

3.1 Correlation and convolution (10 points)

Image filtering is a fundamental tool of image processing. Refer to the lecture materials or Chapter 3 of Szeliski's *Computer Vision: Algorithms and Applications* (2nd edition) for further reading.

Implement `cross_correlation_2d` without flipping the kernel.

Implement `convolve_2d` by flipping both kernel axes and calling your correlation function.

3.2 Gaussian blur and low-pass filtering (10 points)

Low pass filters are used to remove high-frequency details from an image, making it appear smoother. You can create a low pass filter by applying a Gaussian blur, which computes a weighted average of neighboring pixels.

Implement `gaussian_blur_kernel_2d` for a given odd size k and positive standard deviation σ.

Implement `low_pass` using this kernel and your convolution function.

3.3 Subsampling (5 points)

Subsampling reduces the resolution of an image by discarding rows and columns. Given an image of dimensions $M \times N$, subsampling by a factor of s will result in a new image with dimensions $(M/s) \times (N/s)$, where M, N should be divisible by s.

Implement a simple downsampling method: discard every alternate row and column to produce an image that is half the size of the original.

3.4 Gaussian pyramid (20 points)

A Gaussian pyramid is a multi-scale representation of an image, where each level is progressively blurred and downsampled. At each level of the pyramid, the image is smoothed using a Gaussian filter and then downsampled to reduce its resolution.

Implement a four-level pyramid: the original image and images at approximately 1/2, 1/4, and 1/8 of its width and height.

3.5 Filtered vs direct subsampling (10 points)

Use a textured image as the input to the Gaussian pyramid in the previous part. From the same input, build a second four-level pyramid by directly downsampling each level by a factor of two, without Gaussian filtering or other smoothing. Compare the corresponding levels of the two pyramids. Report whether they differ, describe any visible differences, and explain the reasons, including possible aliasing.

Requirements

1. **Report.** Answer Questions 1 and 2 and include the pyramid comparison from Question 3. You may prepare the report in LaTeX, Markdown, Word, or as scanned handwritten pages, but please do not answer directly on this file. Submit the final report as a PDF.

2. **Code.** Submit `gaussian_pyramid.py` with all required functions. Save both pyramids at 1/2, 1/4, and 1/8 resolution. Use relative paths for all image input and output. Briefly list the dependencies and how to run the script. Do not use library functions that directly perform the required filtering or pyramid operations. NumPy array operations are allowed; PIL, OpenCV, or matplotlib may be used for image input, output, and display.

3. **AI declaration.** In the report, briefly describe any AI assistance and its purpose. If none was used, say so. If undeclared AI-generated content is found and contains factual errors or code that cannot run, the affected part will receive zero points.

4. **Submitting.** Place the PDF report, script, and images in one folder. Zip the folder with the name YOURNAME_YOURID_HW1.zip. For example, 金雨桐_2300013101_HW1.zip.